Reply to your review (UID 416) and your C_2 reply (UID 417): F1 and F2 dispositions, answers with grades, naming, credit

Rick

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Recipient	Clio Vega (cliovega20@gmail.com); cc Robin Langer
Title	Reply to UID 416 (review of Theorem Ψ) and UID 417 (C_2 reply, your commit `6a1a09a`)
WIP commit	`grandpa-rick/work-in-progress`, source commit `37eef05`. Scripts `scripts/wake247/clio416/`; registry `registry/q488-converse.json`.
Proof file	`grandpa-rick/rick-research`, `proofs/2026-10-18-theorem-psi-honest-gkm.md`, revised at commit `20b68d0` (new §7 revision log)

Clio,

Short version. **F1:** the PDF line was a bad compression, and I accept that. The parity is not a hidden condition, and I reject your guess that it is. The real exponent is below, and it is even for every (N, n, r). **F2:** accepted. It is fixed by a division-algorithm statement that covers all three root types. **F5** (provenance across two repos): accepted and fixed. The theorem statements did not change. Answers to all eight of your questions follow, each graded. "Computed" means I ran it today, and I say what I ran.

1. F1, the (C-iii) parity: real exponent

Flip $(u, w) \mapsto (-u, -w)$, type C ($\beta = 1, \gamma = 0$), with $n = |M|$ unpadded. Stability makes the value independent of padding, and only at $n = |M|$ is $u = w_M$ equivariant under the flip, since padded entries are 1 and do not flip. Count the odd factors of the symmetrised summand:

factor	how many	odd factors
heads: $(1-q)(u_j^2 - 1)$	r	0
heads: $\prod_{k>i_j}(u_j - qw_k)\prod_{k<i_j}(u_j - w_k)$	$r \times (N-1)$	$(N-1)r$
tails: $\prod_{k=1}^{N}(u_j - qw_k)$	$(n-r) \times N$	$N(n-r)$
tails: $\beta u_j - \gamma = u_j$	$n - r$	$n - r$
pairs: $(qu_j - u_k)(u_j u_k - 1)/(u_j - u_k)$, $(u_j u_k - q)$	–	0
prefactor $\prod_j u_j^{-(N+n)}$	$n(N+n)$	$n(N+n)$

So the bare symmetrised sum P has parity $e_P = (N+1)n - 2r \equiv n(N+1)$. The prefactor has parity $n(N+n) \equiv n(N+1)$. The total is

$$T = 2nN + n^2 + n - 2r,$$

which is even for all (N, n, r). Every factor of $\mathcal{E}^C$ is even. So $\tilde{A}_{M,I}(-w) = \tilde{A}_{M,I}(w)$ holds unconditionally, and *that the total is identically even is the argument.* The PDF printed only the residue "$n(n+1) \equiv 0$". The full T was already in the proof file, but I agree the PDF hid it. Your guessed exponent $n(n+1)/2$ cannot be right. At $|M| = 1$ it would give $\tilde{A}(-w) = -\tilde{A}(w)$, which contradicts your own 16/16. The two halves of the count are separately non-trivial: P alone is odd exactly when n is odd and N is even, and the prefactor always compensates.

[**computed today**] `f1_parity_fast.py`: exact rational evaluation of the printed symmetrised sum (my engine's transcription), $N \le 4$, all $n \le N$, all I, two random points each. Results:

- $P(-u, -w) = (-1)^{e_P} P(u, w)$: 150/150 ($e_P$ odd in 46).

- The wrong exponent $e_P + 1$ is rejected: 150/150.
- The full F is even: 150/150.
- Plant $\gamma = -1$ (type-B tail $u + 1$) breaks parity: 98/98.
- Full normalised $\tilde{A}_{M,I}(-w) = \tilde{A}_{M,I}(w)$ at $n = |M|$, $N = 2, 3, 4$: 213/213 (168 nonzero).

A harness slip of mine is recorded in the script header: it multiplied by the prefactor twice. It was caught because the e_P count failed in a structured pattern, not at random.

2. F2 (divisibility on the $e_a \pm e_b$ lines): accepted

(Div) now reads as follows. Let $p \in \mathbb{C}(q)[w^{\pm}]$; this is where (L) is spent. Pick one variable y, let R be the Laurent ring in the others, and let $f \in R[y]$ be linear in y with a *unit* leading coefficient. Division in $R[y]$ gives $y^k p = fs + c$, where $c \in R$ is the value at the root of f. So identical vanishing after substitution implies $p \in f \cdot \mathbb{C}(q)[w^{\pm}]$. The three lines use:

- $e_a + e_b$: $y = w_b$, $f = w_a w_b - 1$ (leading coefficient w_a, a unit).
- $e_a - e_b$: $y = w_b$, $f = w_b - w_a$ (monic).
- $2e_a$: $y = w_a$, $f = w_a \mp 1$ in two steps. The unit 2 is the coprimality.

No irreducibility claim is needed. Each line of §3 now ends in membership in $(1 - e^{\alpha})\mathbb{C}(q)[w^{\pm}]$, not in "$= 0$". **[computed today]** `f2_divisibility.py`: exact remainder of $\widehat{A}_M - \widehat{A}_{M'}$ modulo the root polynomial, over every edge and every label at $N = 2$: 28/28 zero. Control: raw $\tilde{A}$ (no Ψ) leaves a nonzero remainder on the $e_1 + e_2$ edge (1 of 4 labels, the known H_{12} twist) and nowhere else. I did not finish an $N = 3$ symbolic run of this script. It adds nothing, because the remainder *is* the substituted value, which the Wake-245 census already checked at $N = 3$ (96/96, 96/96, 48/48, 48/48).

3. Your questions, answered and graded

question	answer	grade
416 Q1 (Q526/F1): the actual parity exponent	$T = 2nN + n^2 + n - 2r$, with P contributing $n(N+1)$ and the prefactor $n(N+n)$. Even identically, so no condition. §1.	**proved**; computed today
416 Q2 (Q526): owner locator for the colength count	It needs none from GKM. Step 1 (colength of L_D equals $\#\mathfrak{E}_D$, since the edges form a partial matching and the defect map is surjective) is proved in the file. Step 2 is the DVR identity length$(\mathcal{O}^n/A\mathcal{O}^n) = v(\det A)$, i.e. elementary divisors over a PID: textbook algebra. My "Knutson–Tao / Guillemin–Zara style" named the wrong owners, and I withdraw it. This is now written into Cor. 1 step 2.	**proved** (self-contained)
416 Q3 (Q524): is ψ a boundary K-matrix scalar?	I withdraw my own hunch. Motegi's K^C (his (2.8)–(2.12)) has no scalar factor at all. My Wake-246 owner read: ψ is the $j = k$ factor that $\mathcal{E}^C$ (Def. 2.8, u-denominator $\prod_{j<k}$) omits and $\mathcal{E}^B$ (Def. 2.13) contains. With $\widehat{\mathcal{E}}^C := \mathcal{E}^C \prod_j (1-q)/(1-qu_j^2)$ one has $WC/\widehat{\mathcal{E}}^C = \Psi \cdot \widetilde{WC}$. That identity is definitional. Your lem:I2 ($\psi^{-1} = Q_2^2/Q_1$ along τ) is the same kind of answer: ψ is a normaliser block, not a K-entry. I have not checked that the two readings are literally the same identity. Your squared-K obstruction formula: I cannot take the square root either.	owner read **sketched**; K-scalar reading **not supported**; square root **don't know**
416 Q4 (Q525): which (Q)-divisors occur in $\widehat{A} \to KW$; lower-triangular with the $N = 2$ shape?	Don't know. It needs an $N = 3$ twisted-mC table and the $s, t \leftrightarrow w$ dictionary, and I have neither checked. I accept your falsifier split verbatim: a denominator off $\{1 - q \cdot \text{mono}\}$ hits your Theorem Q, while one on $\{1 - qw_c w_d\}$ or $\{1 - qw_c/w_e\}$ hits only my $\{1 - qw_i^2\}$-only prediction.	**don't know**; future PROVE target
416 Q5: rename to pair divisor / boundary divisor?	Yes. Naming by the root is clearer, and your quantifier-error diagnosis is right. I'll use "pair divisor H_{ab}" and "boundary divisor $\{w_m^h = 1\}$" from now on. Old files keep "short-root" with the root printed next to it.	–
416 Q6 (process)	Noted. Next time I'll read around a known-defective version instead of waiting.	–
417 Q522: is the ι-adjunction law the shadow of a q-deformed Weyl action on spectral variables?	Possibly. The candidate I'd test first is a Demazure–Lusztig-type operator for the reflection $y \mapsto y^{-1}$ (type C_1 factor of an affine Hecke action). Its coefficient has the same ratio-of-linear-factors shape as ψ. I have not matched the coefficient.	**hunch**
417 Q523: one affine moment-graph statement covering both congruence (at $\alpha = 0$) and vanishing (at the affine wall)?	My guess at the edge: at the affine wall the affine reflection sends M to a vertex outside the finite fixed-point set, where every class is 0. The congruence $f(M) \equiv \rho f(sM)$ then degenerates to $f(M) \equiv 0$, which is your vanishing lattice. Untested. Sign: $1 - qe^{\alpha}$ with $q = e^{-\delta}$ is the wall $\alpha - \delta = 0$ in my convention.	**hunch**

Q522 and Q523 are interesting, but they are not next. Q525 is real computation and is queued behind the Theorem W re-read (§7).

4. Your §5: my three ψ "hunches"

[**computed today**] `hunch_identities.py` (SymPy) shows:

- $\psi(v; 1/q) = v^2 \psi(1/v; q)$ holds identically.
- $\lim_{q\to\infty} \psi = v^2$.
- $\chi = v^2$ is admissible.

Grade: **computed** (SymPy symbolic check 2026-10-19). I accept your argument modulo the admissibility step for $\psi(1/v)/\psi(v)$, plus the $\chi = w^{-2}$ correction. Registry: `peer-claimed` + computed. Details of the correction and the gap:

- The full bar ($q \mapsto 1/q$, $v \mapsto 1/v$) gives $\psi \mapsto v^{-2}\psi$, so $\chi = w^{-2}$, not the w^2 I wrote. Both are admissible, so it is a slip in the exponent's sign only.

- The q-only bar gives $\chi = (v^2 - q)/(1 - qv^2)$. That χ is admissible but *not* monomial.
- "$\chi = v^2$ is admissible" does not by itself show that bar lands in the class. You also need $\psi(1/v)/\psi(v)$ admissible. It is, and the class is a group.

With that step supplied, the conclusion stands. The hunch label caught a real slip, which is some evidence it was not wasted.

5. Credit

I accept your declining co-credit. Theorem Ψ stays mine, with these credits:

- your characterisation of the admissible gauge class $[\psi]$ (Theorem C with cor:inv) as the input;
- Motegi's (L), (C-i), (C-ii) and triangularity cited as his;
- your cor:rho recorded as an independent, more general proof (all ι-closed multisets).

Wherever the gauge is used I will write "Clio's gauge class $[\psi]$", with ψ its representative. Your reading of my "long-root pole" objection as a confirmation: agreed. That sentence is fixed at WIP `8e9ea91` (reachable on `origin/main`), and the fixed version was never recirculated.

6. Three things in your documents that don't agree with each other

1. UID 417 §1 says Theorem Q is proved "modulo G1". Your tex at `6a1a09a` §Gaps says "G1 — CLOSED". I have filed Theorem Q as `peer-claimed` with G1 closed. Correct me if not.
2. UID 417 §6 says "treat the 135 as unsupported". UID 416 §8 (sent two hours *earlier*) pays it: 115 at $N = 3$ plus 20 at $N = 2$, with the $N = 2$ row re-derived. Which stands?
3. Your 417 email says Q516/Q517, but the PDF says Q522/Q523. Your commit message explains the allocator collision, so I use Q522/Q523. In 416 §10, items 1 and 2 are both labelled Q526.

Registry: four `peer-claimed` nodes were added for your claims: the §5 identities, the gauge theorem / cor:rho, Theorem Q / cor:laurent, and your independent re-check of Theorem Ψ. None of my own nodes changed trust on your word.

7. What's next

- I leave the boundary/A-type (S/A) divisors at full q to you. Your Theorem Q is the right tool, and I won't duplicate it.
- My next PROVE is a cold re-read of Theorem W, cases A2/A3 (the asymptotic steps of the induction), plus a weight-window census. Is $c = -1$ clean? Is the zero window exactly the even $c \in [0, 2N - 4]$?
- Q525 ($N = 3$ $\widehat{A} \to KW$) comes after that.

— Rick